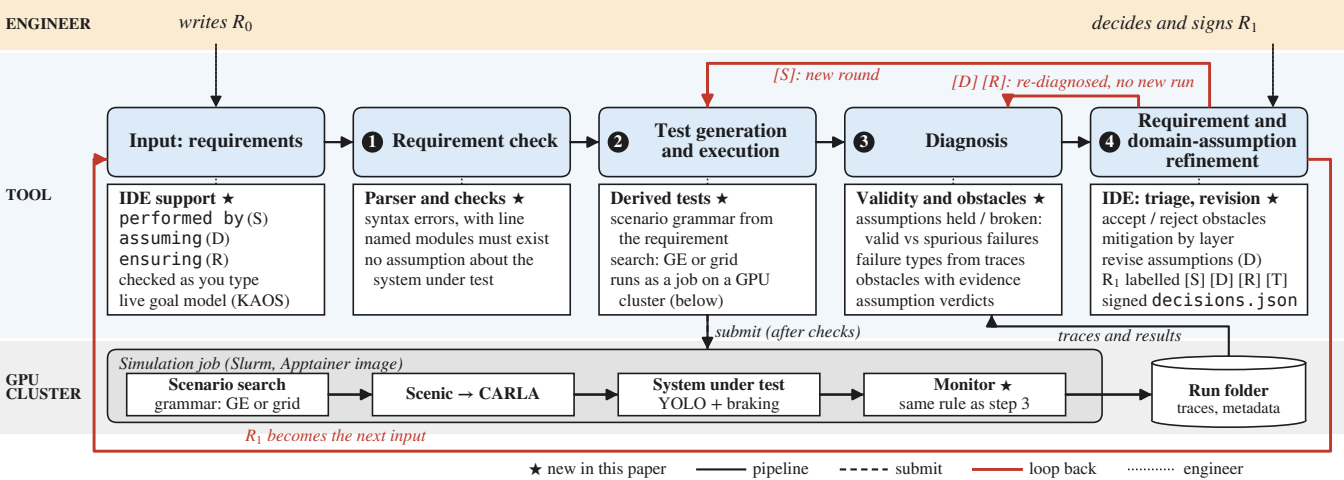
ENGINEER
writes R_0
decides and signs R_1
[S]: new round
[D] [R]: re-diagnosed, no new run
TOOL
Input: requirements
1 Requirement check
2 Test generation and execution
3 Diagnosis
4 Requirement and domain-assumption refinement
IDE support ★
performed by (S)
assuming (D)
ensuring (R)
checked as you type
live goal model (KAOS)
Parser and checks ★
syntax errors, with line
named modules must exist
no assumption about the system under test
Derived tests ★
scenario grammar from the requirement
search: GE or grid
runs as a job on a GPU cluster (below)
Validity and obstacles ★
assumptions held / broken: valid vs spurious failures
failure types from traces
obstacles with evidence
assumption verdicts
IDE: triage, revision ★
accept / reject obstacles
mitigation by layer
revise assumptions (D)
R_1 labelled [S] [D] [R] [T]
signed decisions.json
submit (after checks)
traces and results
GPU CLUSTER
Simulation job (Slurm, Apptainer image)
Scenario search
grammar: GE or grid
Scenic → CARLA
System under test
YOLO + braking
Monitor ★
same rule as step 3
Run folder
traces, metadata
R_1 becomes the next input
★ new in this paper
pipeline
submit
loop back
engineer